

Peer skill identification and social class

Evidence from a field experiment

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16 December 2024

Inequality and social class

- Persistent differences in labor market outcomes based on social class that go beyond race and gender
- Social capital stands out as a key factor controlling for productivity and demographics (Stansbury and Rodriguez, 2024)
- Better connected individuals have better employment opportunities (Montgomery, 1991; Calvó-Armengol & Jackson, 2004; Chetty et al., 2022a)
- Biases like homophily may prevent social classes from interacting with one another (McPherson, Smith-Lovin, and Cook; 2001)

This paper

- Field experiment in a highly segregated society with very high income equality
- Study the existence of social class bias in a university setting where we abstract away from spatial segregation between social classes (Chetty et al., 2022b)
- Inspiration from job referral experiments where participants identify and nominate skilled individuals under performance incentives (Beaman & Magruder, 2012; Beaman, Keleher, Magruder; 2018)
- Show how well peers identify different skills of their classmates, and whether there is a bias against lower social classes
- Introduce quota-like incentives as a treatment to mitigate social class bias

Setting in Colombia

- Among the most unequal countries in the world, where the top decile earn 50 times more than the poorest (World Bank, 2024)
- Nationwide classification system which categorizes the population into 6 strata, aligning with existing social class divisions (Guevara & Shields, 2019; Uribe-Mallarino, 2008)
- Sample from end-of-first-year students in a mid-sized university with access to academic transcripts and demographics
- Mixed student body with 57% Low-SES in sample distribution

Design

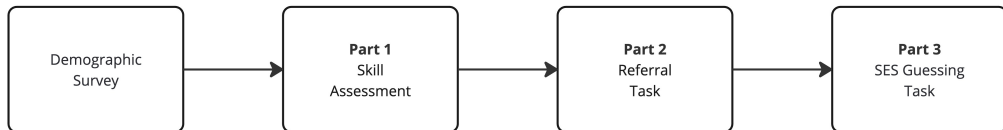
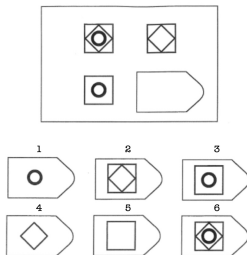


Figure: Timeline

- Three sequential parts with sessions organized at the classroom level
- Experimental variation in Part 2 with incentives to mitigate social class bias

Part 1: Skill Assessment



(a) Cognitive Skill

Fotografía 1: Elige la palabra que mejor describa la fotografía



☐ Nervioso/a

☐ Sarcástico/a

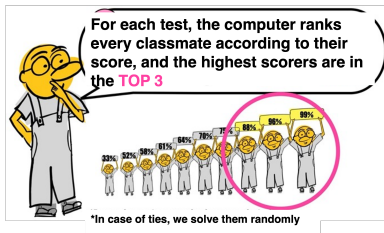
☐ Curioso/a

☐ Amigable

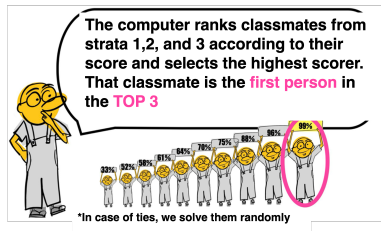
(b) Social Skill

- Participants perform two incentivized skill tests inside the classroom
- No performance feedback, only a brief explanation of the skill
- Uncorrelated and job market relevant (Kuhn & Weinberger, 2005; Heckman, 2006; Heckman & Kautz, 2012; Weinberger, 2014; Deming, 2017)

Part 2: Skill Referrals



(a) Baseline



(b) Treatment

- Participants privately nominate three peers from their classroom for each skill
- Nominations among the top 3 highest performing classmates to earn a bonus
- **Treatment** changes the top 3 composition for half of the classroom
- Testable assumption: Can peers identify different skills?

Part 3: SES Guessing Ability

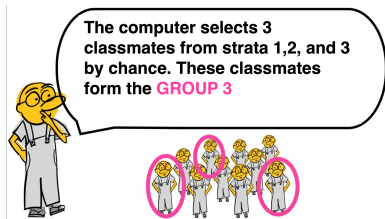


Figure: SES Guessing

- Participants privately nominate three low-SES peers from their classroom
- Nominations among low-SES to earn a bonus
- Testable assumption: Can peers observe social class? distribution

Procedure



Figure: Participants in a classroom

- 863 students across 35 classrooms invited, 713 completed (83%)
- **Link** to pre-registered design and hypotheses
- Median length 20 minutes
- \$26 for 117 lottery winners

Descriptive Statistics

Table: Treatment Balance

Variable	Baseline	Treatment	<i>p</i> -value
Low-SES (%)	59.0	55.0	0.297
Cognitive skill ^a	10.04	10.27	0.322
Social skill ^b	18.45	18.50	0.886
GPA	3.95	3.95	0.828
# semesters at UNAB	3.17	3.18	0.916

^a Max. score: 18 points

^b Max. score: 36 points

Note: *p*-values from two-sided *t*-tests comparing group means.

Dependent Variable

$$y_{ic}^s = \frac{\sum_{j \neq i} r_{ijc}^s}{n_c - \mathbb{1}(i \in B_c)} \times 100$$

- Percentage share of nominations individual i gets for skill s
- From classmates in class c and in Baseline (B_c)
- Sum of nominations ($\sum_{j \neq i} r_{ijc}^s$) divided by the max. possible nominations (n_c)
- Accounting for no-self-referral rule
- Allows comparing characteristics of individuals who get nominated more often across classrooms of different sizes

Dependent Variable

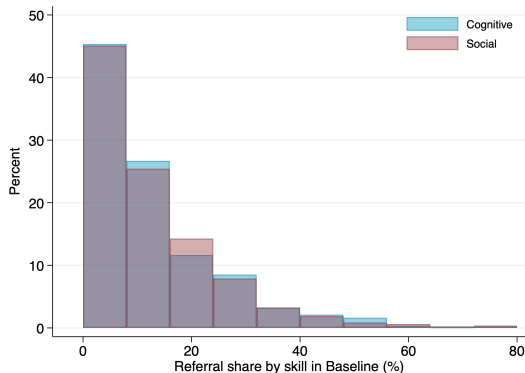


Figure: Distribution of nominations in Baseline

Result 1: Can peers identify skills at baseline?

	(1) Cognitive	(2) Social
Skill	1.197** (0.479)	0.037 (0.510)
Constant	12.986*** (0.653)	13.049*** (0.521)
Observations	665	665

Table: Share of nominations conditional on skill

- Cognitive skill can be identified by peers, but not social skill
- Magnitude of the effect very modest

Result 1: Can peers identify skills at baseline?

	(1) Cognitive	(2) Social
Skill	0.873* (0.467)	-0.278 (0.460)
GPA	3.949*** (0.664)	3.429*** (0.581)
Constant	12.806*** (0.662)	12.891*** (0.636)
Observations	665	665

Table: Share of nominations conditional on skill and GPA

- Participants use academic performance as a proxy for both skills

Result 2: Nomination strategies

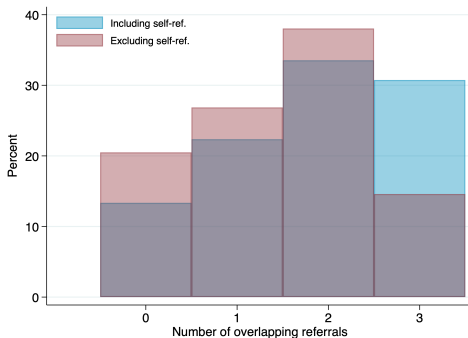


Figure: Distribution of identical nominations

- Up to 6 unique nominations can be made but the observed diversity is much less
- More than half of the sample nominate the same classmates 2 or 3 times

Result 2: Nomination strategies

$$y_{ic}^s = y_{ic}^{s,single} + y_{ic}^{twice}$$

- Percentage share of nominations received can be decomposed into nominations for skill s only versus for both
- Help us identify whether nominations for one skill are different than for both

Result 2: Nomination strategies

	(1) Twice	(2) Cognitive Only	(3) Social Only
GPA	3.172*** (0.464)	0.801** (0.391)	0.260 (0.334)
Cognitive Skill	-0.042 (0.416)	1.006*** (0.270)	
Social Skill	-0.353 (0.304)		0.086 (0.381)
Constant	7.407*** (0.524)	5.400*** (0.386)	5.485*** (0.376)
Observations	665	665	665

- 3 types: GPA, GPA and cognitive skill, neither social skill or GPA

Result 3: Treatment and social class

	(1) Cognitive	(2) Social
Treat	-0.073 (0.755)	0.299 (0.716)
Low-SES	-1.230 (1.079)	0.364 (1.282)
Treat $\times$ Low-SES	-0.167 (1.117)	-0.835 (1.181)
Constant	13.551*** (0.986)	12.706*** (0.952)
Performance controls	YES	YES
Observations	1,330	1,330

- No low-SES bias when controlling for skills and GPA, no treatment effect

Result 3: Treatment and social class

	(1) Twice	(2) Cognitive Only	(3) Social Only
Treat	0.436 (0.817)	-0.509 (0.598)	-0.136 (0.523)
Low-SES	0.857 (0.920)	-2.074*** (0.722)	-0.510 (0.613)
Treat $\times$ Low-SES	-1.584 (1.159)	1.417** (0.656)	0.750 (0.717)
Constant	6.952*** (0.727)	6.591*** (0.654)	5.765*** (0.562)
Performance controls	YES	YES	YES
Observations	1,330	1,330	1,330

- Low-SES bias only for cognitive skill nominations, treatment mitigates

Result 3: Treatment and social class

	(1) Low-SES	(2) High-SES
Treat	0.541 (0.754)	-0.872 (0.946)
Low-SES	-0.724 (0.717)	-1.681 (0.993)
Treat $\times$ Low-SES	-0.552 (0.927)	2.711** (1.289)
Constant	4.359*** (0.472)	4.766*** (0.663)
Performance controls	YES	YES
Observations	1,328	1,310

- High-SES drive the results when adjusting for low-SES share across classrooms

Conclusion

- Limited peer capacity in identifying cognitive and social skills: First-year students may not know each other's skills very well, or the choice/design of the skill assessment was not optimal
- For half of the sample, nomination decisions are characterized by grades instead
- No general social class bias in nominations and no treatment effect when controlling for performance
- Less than 25% of nominations affected by low-SES bias, driven by high-SES peers
- For this subsample treatment is effective as well

Distribution of SES guessing ability

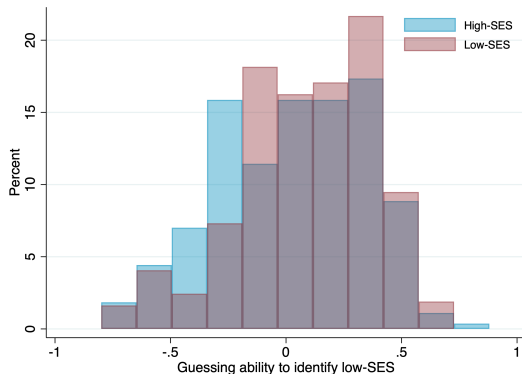


Figure: SES guessing ability [back](#)

Distribution of strata

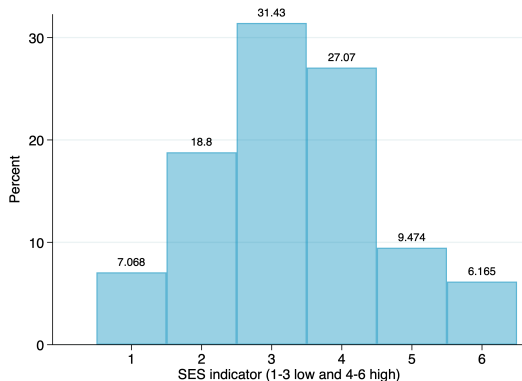


Figure: Sample strata distribution at UNAB [back](#)